PROBABILITY

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This is an important topic for MBA entrance tests. Recent trends show that this importance is increasing. Besides the entrance tests, the topic is an important part of the management courses themselves. Therefore, students aspiring to be future managers need a sound understanding of the basics.

Natural phenomena are of two types - deterministic and probabilistic. For example, the direction in which or the time at which the sun rises everyday is a deterministic phenomenon, while, where or when it may rain is a probabilistic phenomenon. Similarly, **experiments** (the operations of doing or observing something resulting in some final outcomes) are of two types - those in which the outcome is definite and others in which the actual outcome may be any one of the many possible outcomes. For example, hydrogen is allowed to react with oxygen. They react in a certain ratio and produce water. The outcome is definite-while if a coin is tossed, it may turn up showing either heads or tails, the outcome may be one of the two possible outcomes. Experiments of the second kind are called **random experiments** (RE). One instance of such an experiment is a **trial**. The set of all possible outcomes for a particular random experiment is its **sample space** S. This corresponds to the concept of the universal set in set theory. Each outcome is said to be a point in S. Thus for the experiment of tossing a coin, the sample space is the set of the two outcomes - heads and tails. For the experiment of rolling a die, it is {1, 2, 3, 4, 5, 6}. For drawing a card from a deck of cards it is the set of all the 52 possible outcomes - corresponding to the 52 cards. Any subset of the sample space is a simple event. Two (or more) events which occur for two (or more) different experiments - or for two (or more) trials of the same experiment are called **compound events**. Thus if a coin is tossed and a die is rolled, the event of getting a head (in the case of the coin) and say 5 (in the case of the die) is a compound event. Similarly, if a die is rolled twice, the event of getting an even number on the first roll and an odd on the second is a compound event.

Equally likely outcomes:

If a normal coin is tossed, it may come up either heads or tails. Both the outcomes are equally likely. We can accept this intuitively, even though at the moment, we do not know how to compute (or measure) the probability of either outcome. For the purpose of this discussion, we start with this assumption - that we can recognise intuitively whether all the possible outcomes are equally likely or not. We know from experience that all coins are not normal. Sometimes, the mass in the coin is so distributed that it shows up one side more than the other. Such coins (or dice) are said to be biased.
We can now consider one definition of probability. This is the only one that we need for the questions that we shall face. For a random experiment with n "equally likely" outcomes, if E is an event which can be considered to have occurred for m of the outcomes, the probability (mathematical probability or a priori probability) of E is $\frac{m}{n}$, i.e. $P(E) = \frac{m}{n}$

The **complement** of an event is the event of the non-occurrence of E. It is denoted by $\bar{E}$ and $P(\bar{E}) = 1 - \frac{m}{n}$.
For example, if the RE is tossing a coin and E is the event of getting heads, $P(E) = \frac{1}{2}$. Also $P(\bar{E}) = 1 - \frac{1}{2} = \frac{1}{2}$.
The complement of getting a head is getting a tail. If the RE is rolling a die and E = {2, 3}, $\bar{E}$ = {1, 4, 5, 6}. In this case, $P(E) = \frac{1}{3}$, $P(\bar{E}) = \frac{2}{3}$

With this definition, we can consider the two extreme cases. An event is any subset of S. If E is the null set P(E) = 0 (an impossible event) and if E = S, P(E) = 1 (a certain event) For example, let the RE be rolling a die and consider the "event" of getting a 0. In our notation, E would be the null set. For no element of S, can it be said that E has occurred. $\bar{E}$ would be the event of not getting a 0 and $P(\bar{E}) = 1$

Instead of E saying that the probability of an event is m/n, we can also say that the odds in favour of the event are m to n – m i.e $P(E)/P(\bar{E})$. Similarly, the odds against the event are n – m to m i.e. $P(\bar{E})/P(E)$.

Mutually Exclusive Events:

If there is a set of events, such that if any one of them occurs, none of the other can occur, the events are said to be mutually exclusive. Consider the RE of rolling a die and the following events.
$E_1 = \{1\}$
$E_2 = \{2, 3\}$
$E_3 = \{4, 5\}$
The three events are mutually exclusive.

Collectively Exhaustive Events:

If there is a set of events such that at least one of them is bound to occur, the events are said to be collectively exhaustive. For the RE considered above, if $E_4 = \{1, 2, 3, 4\}$ and $E_5 = \{3, 4, 5, 6\}$, E_4 and E_5 are collectively exhaustive.

If a set of events are both mutually exclusive and collectively exhaustive, the sum of their probabilities is 1.

Addition Theorem of Probability:

If A and B are two events, then
$P(A \cup B) = P(A) + P(B) - P(A \cap B)$
This result follows from the corresponding result in set theory. If n (X) represents the number of elements in set X, $n(X \cup Y) = n(X) + n(Y) - n(X \cap Y)$

Example: If a die is rolled, what is the probability that the number that comes up is either even or prime?
A = The event of getting an even number = {2, 4, 6}
B = The event of getting a prime = {2, 3, 5}
$A \cup B = \{2, 3, 4, 5, 6\}$
$A \cap B = \{2\}$
$P(A) = \frac{3}{6}$, $P(B) = \frac{3}{6}$, $P(A \cup B) = \frac{5}{6}$ and $P(A \cap B) = \frac{1}{6}$.
We can verify that
$P(A \cup B) = P(A) + P(B) - P(A \cap B)$

Conditional Probability and the Multiplication Theorem of Probability

The events that we have considered so far are without reference to other events (or conditions). But, very often, we need to consider events, in relation to other events (or conditions). We can continue with the same RE. Let A be the event of getting a prime and B be the event of getting an even number, i.e. A = {2, 3, 5} and B = {2, 4, 6}, P (A) = $\frac{3}{6}$

But if B is known to have occurred, then P (A) = $\frac{1}{3}$.

We write this as follows:

$P\left(\frac{A}{B}\right) = \frac{1}{3}$, This is read as follows:

The probability of A, given B is $\frac{1}{3}$.

In general,

$$P\left(\frac{A}{B}\right) = \frac{P(A \cap B)}{P(B)} \text{ or } P(A \cap B) = P(B).P\left(\frac{A}{B}\right) \rightarrow (1)$$

Also, as $P\left(\frac{B}{A}\right) = \frac{P(B \cap A)}{P(A)}$, it follows that

$$P(B \cap A) = P(A).P\left(\frac{B}{A}\right). \quad \rightarrow \quad (2)$$

This result (1) or (2) is known as the multiplication theorem of probability.

If $P\left(\frac{A}{B}\right)$ = P (A), then A and B are said to be independent.

A is independent of B, because whether B occurs or does not occur, the probability of A does not change.
Example, A number is selected at random from the integers 1 to 50. A is the event of getting a multiple of 5 and B is the event of getting an even number.
Then A = {5, 10, 15....45, 50}, B = {2, 4, ..., 48, 50}

$P(A) = \frac{10}{50}$

$P\left(\frac{A}{B}\right) = \frac{5}{25}$, also $P\left(\frac{A}{\overline{B}}\right) = \frac{5}{25}$

$\therefore$ A and B are independent events.

If A and B are independent, $P\left(\frac{A}{B}\right) = P(A)$

$$\Rightarrow \frac{P(A \cap B)}{P(B)} = P(A) \qquad \text{or } P(A \cap B) = P(A).P(B)$$

If A and B are independent, so are A, $\overline{B}$; $\overline{A}$, B and $\overline{A}, \overline{B}$

i.e. if $P(A) = P\left(\frac{A}{B}\right)$ then each of the following is true

$$P(A) = P\left(\frac{A}{\overline{B}}\right)$$

$$P(B) = P\left(\frac{B}{A}\right) = P\left(\frac{B}{\overline{A}}\right)$$

We can show the following results

If $P(A) = P\left(\frac{A}{B}\right)$, then $P(A) = \left(\frac{A}{\overline{B}}\right)$.

Also $P\left(\frac{B}{A}\right) = P(B) = P\left(\frac{B}{\overline{A}}\right)$

Pairwise independence and mutual independence:

If A, B, C are three events such that each of the 3 pairs A, B; B, C and C, A are independent. A, B, C are said to be pairwise independent. Let P(A), P(B), P(C) be a, b, c respectively.
As A, B are independent, P $(A \cap B)$ = ab,
As B, C are independent, P $(B \cap C)$ = bc,
As A, C are independent, P $(C \cap A)$ = ca

Even if these three conditions are true, P $(A \cap B \cap C)$ is not necessarily equal to abc. In case it is, the events are said to be mutually independent. We note that mutual independence is a stronger condition. It implies pairwise independence (while pairwise independence does not necessarily mean mutual independence)

For more events, we can generalise the concept, we can talk of pairwise (or 2-wise), tripletwise (or 3-wise), quadrapletwise (or 4-wise) independence and so on. Mutual independence of n events would mean i-wise independence for i = 2, 3, 4....n

Example:

A positive integer from 1 to 60 is selected at random.
A is the event of selecting a multiple of 3.
B is the event of selecting a multiple of 4.
C is the event of selecting a multiple of 5.
The following results are true.

$a = P(A) = \frac{20}{60} = \frac{1}{3}$	$P(A \cap B) = \frac{5}{60}$	
$b = P(B) = \frac{15}{60} = \frac{1}{4}$	$P(A \cap C) = \frac{4}{60}$	$P(A \cap B \cap C) = \frac{1}{60}$
$c = P(C) = \frac{12}{60} = \frac{1}{5}$	$P(B \cap C) = \frac{3}{60}$	

Thus, P $(A \cap B)$ = ab, P $(A \cap C)$ = ac, P $(B \cap C)$ = bc and P $(A \cap B \cap C)$ = abc,
$\therefore$ A, B, C are not merely pairwise independent but also mutually independent

Bayes' Theorem:

Let $A_1, A_2 \ldots. A_n$ be a mutually exclusive and collectively exhaustive events with respective probabilities of p_1, $p_2 \ldots\ldots p_n$. Let B be an event such that P (B) $\neq$ 0 and $P\left(\frac{B}{A_i}\right)$ for i = 1 to n be $q_1, q_2 \ldots. q_n$. Then the conditional probability of A_i given B is $\frac{p_i q_i}{p_1 q_1 + p_2 q_2 + \ldots. + p_n q_n}$

Example:

Box 1 contains 3 white and 2 black balls. Box 2 contains 1 white and 4 black balls. A ball is picked from one of the two boxes. It turns out to be black. Find the probability that it was drawn from box 1.

The data is tabulated below

	Box 1	Box 2
White	3	1
Black	2	4

The event that box 1 is selected is say A_1.
The event that box 2 is selected is A_2

$\therefore\ p_1 = P(A_1) = \frac{1}{2}$ and $p_2 = P(A_2) = \frac{1}{2}$

Let B be the event that a black ball is selected

$\therefore P\left(\frac{B}{A_1}\right) = \frac{2}{5}$ and $P\left(\frac{B}{A_2}\right) = \frac{4}{5}$

Now it is given that a black ball has been drawn and we need to find P (A_1) i.e. we need $P\left(\frac{A_1}{B}\right)$. We use the result above

$$P\left(\frac{A_1}{B}\right) = \frac{P(A_1)P\left(\frac{B}{A_1}\right)}{P(A_1)P\left(\frac{B}{A_1}\right) + P(A_2)P\left(\frac{B}{A_2}\right)}$$

$$= \frac{\left(\frac{1}{2}\right)\left(\frac{2}{5}\right)}{\left(\frac{1}{2}\right)\left(\frac{2}{5}\right) + \left(\frac{1}{2}\right)\left(\frac{2}{5}\right)} = \frac{2}{2+4} = \frac{1}{3}$$

Expected value

The concept of "expected value" is very important in the Theory of Probability. This concept is very useful in managerial decision-making.

The theory of probability has its origin in gambling. When people went to gambling houses or casinos where they used to get certain money if they achieved a certain result in the game. Mathematicians wanted to find out as to how much a person will earn if the game is played a large number of times.

Let us say that a man is playing a game of "throwing a die". He is given ₹6 if he throws a "four" and ₹9 if he throws a "six" on the die and not paid anything if he throws any other number (of course, he will have to pay some amount to the gambling house owner each time he wants to throw the die and this aspect will be considered later). Suppose he throws the die a large number of times - say 6,00,000 times. As the number of times the experiment is repeated becomes very large, we know that the number of times each event will occur is given by probability.

A "four" will appear with a probability of 1/6, i.e., it is expected to appear 1,00,000 times out of a total of 6,00,000 times the die is thrown. Similarly, a "six" will appear 1,00,000 times (because the probability is 1/6). Hence, the amount he will get in the long run will be 1,00,000 $\times$ 6 + 1,00,000 $\times$ 9 = 15,00,000. The amount he gets per throw will be 1500000/600000 = ₹2.5.

We say that the person's **expected value** of this game per throw in the long run is ₹2.5.

This can be calculated without the number of throws coming into the picture. Once the events are defined, we should have the probabilities of all the events and the monetary value associated with each event (i.e., how much money is earned or given away if that particular event occurs). Then,

$$\text{Expected Value} = \sum_i [\text{Probability } (E_i)] \times [\text{Monetary value associated with event } E_i]$$

Expected value:

If each outcome of a RE is associated with a value of a variable X, the expected value of X, E = $\frac{\Sigma P_i X_i}{\Sigma P_i}$ = $\Sigma P_i X_i$

Example:

A person tosses a coin. If it comes up heads, he gets ₹10. If it comes up tails, he has to pay ₹5. What is his expected value?

Event	Heads	Tails
Probability	$\frac{1}{2}$	$\frac{1}{2}$
Expectation	10	–5

$$\therefore E = \frac{\left(\frac{1}{2}\right)10 + \left(\frac{1}{2}\right)(-5)}{\left(\frac{1}{2}\right) + \left(\frac{1}{2}\right)} = 2.5$$